Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

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The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Scalable AI-based Wearable Medical Device Platform for Continuous Monitoring and Real-time Cardiorespiratory Disease Diagnosis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (v), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (v), Next-gen Communication ( ), Advanced Robotics & Manufacturing (v), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Digital Healthcare		Health Diagnostics	
	Medium Category	**Small Category**	**Medium Category**	**Small Category**
	Wearable Device	Cardiorespiratory disease detection	Wireless Non-Invasive Device	AI-based disease detection
Concept of Proposed Technology	The proposed technology aims to develop a **soft, AI-enabled wearable stethoscope for continuous cardiorespiratory monitoring** that enables real-time assessment of heart and lung sounds in both clinical and home environments. The system is designed as a skin-conformal, lightweight wearable device that can be comfortably worn for extended periods, overcoming the limitations of conventional rigid and intermittent stethoscopes. By integrating continuous acoustic sensing with AI-based signal analysis, the wearable stethoscope automatically detects abnormal respiratory and cardiac patterns associated with disease progression or			

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	acute exacerbation. The platform further provides intuitive real-time alerts to users, caregivers, or clinicians, enabling early recognition of physiological changes and timely intervention. This technology transforms auscultation from an episodic, clinician-dependent practice into a **continuous, automated, and interactive monitoring process**. Key features: • Soft, skin-conformal wearable stethoscope for continuous physiological sound monitoring • AI-based cardiorespiratory sound analysis for detection of abnormal patterns • Real-time alerting and feedback mechanisms for early awareness and intervention • Wireless data transmission and longitudinal monitoring for clinical and at-home use
Rationale for Joint R&D Project	Continuous cardiorespiratory monitoring using wearable medical devices represents a rapidly growing healthcare market, yet current stethoscope-based solutions remain limited by rigid form factors, intermittent use, and lack of automated interpretation and real-time feedback. Translating a soft, AI-enabled wearable stethoscope into a deployable product requires close integration of advanced sensing algorithms, wearable hardware design, system integration, and manufacturing expertise, which cannot be achieved by a research institute alone. Necessity of Joint R&D: • The research institute provides expertise in wearable sensing, system integration, and AI-based diagnostic algorithms • Korean SME partners provide device hardware development, manufacturing, and patient experimentation capabilities • Close cooperation is required to translate advanced wearable and AI technologies into a deployable medical device that meets clinical, manufacturing, and regulatory requirements Impact of joint R&D: • Promotes the transition from research-level wearable technology to a commercial medical device platform • Enables early detection and management of cardiorespiratory conditions in clinical and home settings • Strengthens the global competitiveness of Korean SMEs in digital healthcare and wearable medical devices
Key Words	1. Soft, Skin-Conformal Medical Device 2. Wearable Wireless Stethoscope 3. Real-time Continuous Cardiorespiratory Monitoring 4. AI-Based Diagnosis 5. State-aware physiological Intervention and Feedback

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objective of this R&D project is to develop a soft, AI-enabled wearable stethoscope that enables continuous and interactive cardiorespiratory monitoring in clinical and home environments. The system integrates wearable acoustic sensing, AI-based diagnosis, and real-time feedback to support early detection and intervention for cardiorespiratory conditions. The specific R&D objectives are: • To develop a soft, skin-conformal wearable sensing system for continuous acquisition of heart and lung sounds • To develop AI-based diagnostic algorithms for real-time detection of abnormal patterns • To implement interactive feedback and intervention mechanisms that enable closed-loop physiological monitoring • To validate system performance in clinical and real-world use scenarios for reliable long-term monitoring			
R&D Target Specifications	The key performance targets of this R&D project are defined as follows: **AI-Based Diagnostic Performance (>95%):** Achieve an abnormal respiratory sound detection accuracy of > 95% for key conditions under real-world monitoring conditions. **Real-Time Alert and Feedback Recognition Performance (> 90%):** Provide distinct alert and feedback patterns corresponding to multiple respiratory conditions (wheeze, crackle, rhonchi, stridor, and cough), and achieve > 90% user recognition accuracy in differentiating alert patterns. **Continuous Monitoring Reliability (>12 hrs):** Maintain stable continuous operation for > 12 hours with wireless data transmission and no critical data loss during wear.			
R&D Plan	This R&D project will be conducted over three years, following a progressive development and validation strategy from device and algorithm development, to clinical and real-world evaluation, and finally to system-level validation and commercialization readiness.			

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	● Year 1: Core Wearable System and AI Algorithm Development • Design and develop a soft, skin-conformal wearable stethoscope for continuous acquisition of cardiorespiratory sounds • Implement signal processing pipelines for stable heart and lung sound extraction under motion and noise • Develop AI-based diagnostic algorithms for detection of abnormal respiratory patterns • Conduct benchtop and pilot laboratory tests to validate sensing performance and diagnostic accuracy ● Year 2: System Integration and Clinical Feasibility Evaluation • Integrate sensing, AI diagnosis, wireless communication, and interactive alert/feedback modules into a unified wearable system • Develop condition-specific alert and feedback patterns for immediate user awareness • Conduct laboratory-scale and clinical feasibility studies (30 patients) to evaluate usability, reliability, and diagnostic performance • Refine hardware and algorithms based on clinical feedback and real-use conditions ● Year 3: Real-World Validation and Technology Advancement • Deploy the integrated system in clinical (50 patients) and at-home monitoring (30 users) environments for extended use • Perform longitudinal studies to assess early detection capability and monitoring robustness • Optimize system stability, user compliance, and data reliability for long-term operation • Evaluate commercialization potential and prepare the technology for industrial transfer with Korean SME partners

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute, BITN group at Georgia Tech, will be responsible for the development of core sensing, AI algorithms, and system-level intelligence for the proposed wearable stethoscope platform. The institute will lead the research on cardiorespiratory signal acquisition, AI-based diagnosis, and interactive feedback mechanisms required for reliable continuous monitoring in clinical and home environments. The specific roles and responsibilities of the research institute include: • Development of wearable sensing methodologies for continuous acquisition of heart and lung sounds • Design of the overall system architecture, including sensing, diagnosis, and feedback integration • Development of AI-based diagnostic algorithms for the detection of abnormal cardiorespiratory patterns • Laboratory-scale validation of sensing performance, diagnostic accuracy, and feedback functionality The research institute will closely collaborate with the Korean SME partner to support advanced manufacturing and fabrication process optimization. This collaboration will align device design with manufacturing constraints and improve fabrication consistency and production quality, supporting the transition toward scalable and reliable wearable medical devices.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for subsystem integration, device deployment, and execution of laboratory and clinical studies required for validation of the proposed wearable stethoscope technology. The SME will play a key role in translating research outcomes into a deployable and manufacturable medical device system, bridging academic research and industrial implementation. The specific roles and responsibilities of the Korean SME partner include: • Subsystem integration of wearable device hardware and wireless communication modules • Deployment and operation of the device in laboratory and clinical environments to support data acquisition • Execution of data collection, labeling, curation, and refinement to support AI-based diagnostic development • Support of system validation through performance testing, reliability assessment, and usability evaluation • Contribution to industry–academia collaboration and workforce development, including training of technical personnel and joint research activities

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to establish a new category of interactive wearable medical devices that move beyond passive monitoring toward real-time diagnosis and action. The proposed wearable stethoscope will address a critical gap in the digital healthcare market by enabling continuous, AI-driven cardiorespiratory monitoring with immediate user feedback in both clinical and home settings. The expected outcomes will include: 1. A market-ready AI-enabled wearable stethoscope platform that demonstrates continuous monitoring and interactive alerting, enabling differentiation from existing passive digital stethoscopes. 2. A scalable product framework that can be extended to additional respiratory and cardiorespiratory conditions, enabling future pipeline expansion in digital healthcare. 3. High-quality clinical and real-world datasets generated through deployment and testing, supporting downstream AI model improvement, regulatory preparation, and future commercial products. In addition, this partnership will promote active collaboration between the research institute and Korean SME partners, enabling mutual learning across wearable sensing, AI-driven diagnostics, and medical device engineering. Through joint development, deployment, and validation activities, the project will contribute to workforce development and training of skilled engineers and researchers with hands-on experience in digital healthcare technologies. The project will support technology leadership in interactive wearable medical devices, positioning Korean SMEs at the forefront of next-generation digital healthcare innovation.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization